

Homework 10: Inner Products

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BCS A, E, F, G

Problem 1. The Frobenius Inner Product or Hilbert–Schmidt Inner Product.

Show that the function $\langle \cdot, \cdot \rangle: M_{m,n}(F) \rightarrow M_{m,n}(F)$ defined by

$$\langle A, B \rangle = \text{tr}(A^*B), \quad \text{for all } A, B \in M_{m,n}(F),$$

where $F = \mathbb{R}$ or $F = \mathbb{C}$, is an inner product on $M_{m,n}(F)$. You can only discuss the case by taking $m = n = 2$ and $F = \mathbb{C}$.

Notation. Here $*$ in A^* denoted the conjugate transpose, also known as Hermitian transpose of $m \times n$ complex matrix A . There are several notations for conjugate transpose, such as A^H or $A^\dagger$ ($\dagger$ reads as dagger and is standard notation used in physics). For real matrices A , the conjugate transpose is just the transpose $A^* = A^T$.

Hint. In other words, this inner product is what we get if we forget about the shape of A and B and just take their dot product as if they were vectors in F^{mn} .

Definition 1. Orthogonality. Suppose V is an inner product space. Two vectors $\mathbf{v}, \mathbf{w} \in V$ are called orthogonal if $\langle \mathbf{v}, \mathbf{w} \rangle = 0$.

Definition 2. Orthonormal Bases. Suppose V is an inner product space with basis $B \subset V$. We say that B is an orthonormal basis of V if

- a) **Mutual Orthogonality:** $\langle \mathbf{v}, \mathbf{w} \rangle = 0$ for all $\mathbf{v} \neq \mathbf{w} \in B$, and
- b) **Normalization:** $\|\mathbf{v}\| = 1, \forall \mathbf{v} \in B$.

Theorem 1. All Inner Products Look Like the Dot Product. If B is an orthonormal basis of a finite-dimensional inner product space V then

$$\langle \mathbf{v}, \mathbf{w} \rangle = [\mathbf{v}]_B \cdot [\mathbf{w}]_B, \quad \text{for all } \mathbf{v}, \mathbf{w} \in V.$$

Corollary 1. The Norm Induced by an Inner Product Looks Like Vector Length. If B is an orthonormal basis of a finite-dimensional inner product space V then

$$\|\mathbf{v}\| = \|[v]_B\|, \quad \forall \mathbf{v} \in V.$$

Elaboration. $\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle} = \sqrt{[\mathbf{v}]_B \cdot [\mathbf{v}]_B} = \|[v]_B\|$.

Problem 2. The Pauli Matrices Form an Orthonormal Basis.

Show that

$$B = \left\{ \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \right\}$$

is an orthonormal basis of $M_2(\mathbb{C})$ with the usual Frobenius inner product.

Theorem 2. Coordinates with Respect to Orthonormal Bases. If $B = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$ is an orthonormal basis of a finite-dimensional inner product space V then

$$[\mathbf{v}]_B = (\langle \mathbf{u}_1, \mathbf{v} \rangle, \langle \mathbf{u}_2, \mathbf{v} \rangle, \dots, \langle \mathbf{u}_n, \mathbf{v} \rangle), \quad \forall \mathbf{v} \in V.$$

Problem 3. Compute $[A]_B$ (the coordinate vector of $A \in M_2(\mathbb{C})$ with respect to B) if

$$B = \left\{ \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \right\}$$

and

$$A = \begin{bmatrix} 5 & 2 - 3i \\ 2 + 3i & -3 \end{bmatrix}.$$

Reference Link. <https://mathworld.wolfram.com/PauliMatrices.html>

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Good Luck